Investing & Compound Growth — Cheat Sheet

Financial Algebra · Unit 6: Investing & Compound Growth

Essential vocabulary

Compound growth
Growth in which earnings are added to the balance and then earn returns themselves

Rate of return
The gain or loss on an investment over a period, expressed as a percent of the starting value

Rule of 72
A shortcut estimating years to double: 72 divided by the annual percent return

Recurring contribution
A regular, repeated deposit into an investment, such as monthly

Future value
The value of money or a series of payments at a future date given a rate of return

Dollar-cost averaging
Investing a fixed dollar amount on a regular schedule regardless of price

Stock
A share of ownership in a company whose value can rise or fall

Bond
A loan to a government or company that pays interest and returns principal at maturity

Index fund
A fund that holds the investments in a market index to match its performance at low cost

Expense ratio
The yearly fee a fund charges, as a percent of the amount invested

Diversification
Spreading money across many investments so that one loss has less impact

Risk
The possibility that actual returns differ from expected returns, including losing money

Volatility

Key formulas

Compound amount	$A = P(1 + \frac{r}{n})^{(nt)}$
Future value of deposits	$FV = PMT \cdot \frac{(1 + i)^N - 1}{i}$
Deposit for a goal	$PMT = \frac{FV \cdot i}{(1 + i)^N - 1}$
Rule of 72	years to double $\approx \frac{72}{\text{rate \%}}$
Percent return	$\frac{\text{end} - \text{start}}{\text{start}}$
Real return	$\frac{1 + \text{nominal}}{1 + \text{inflation}} - 1$
Gain to recover a loss	$\frac{1}{1 - \text{loss}} - 1$

Match the horizon

< 3 years	Savings, CDs
3–10 years	Mix of bonds and stocks
10+ years	Diversified stock-heavy funds
Any horizon	Diversify; keep fees low

Worked example

1. $200/month for 35 years at a hypothetical 7%
2. $i = \frac{0.07}{12}$, $N = 420$
3. FV ≈ $360,000; contributions $84,000
4. Growth ≈ $276,000 — time does most of the work

How much an investment’s value moves up and down over time

Time horizon
How long until the money is needed

Risk tolerance
A person’s ability and willingness to accept investment losses

Real return
Return after adjusting for inflation

Common misconceptions

- High average returns are guaranteed.
- Doubling the rate doubles the result.
- Small fees do not matter.
- Diversification eliminates risk.

Questions to ask before deciding

- When will I need this money?
- What could I lose in a bad year?
- What are the fees?
- Am I diversified?
- Have I built an emergency fund and paid off high-interest debt first?

Decision tips

- Start early, even small.
- Automate contributions.
- Prefer low-cost diversified funds.
- Don’t panic-sell in downturns if your horizon is long.

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